Laboratory 2:
Synthesis and Standard Cell

OBJECTIVES

- The objective of this lab is for students to perform the complete digital design flow from RTL design → synthesis → gate-level simulation using **Genus** and **Xcelium** to carry out the synthesis and verification of a digital circuit.
- Using **Genus**, the RTL design written in HDL is synthesized into a **gate-level netlist** based on a standard cell library, while also generating corresponding timing information in the form of a **Standard Delay Format (SDF)** file and other parameters such as **power, area, and cell usage**.
- Using **Xcelium**, students perform **gate-level simulation** of the synthesized netlist with **SDF annotation**, enabling the verification of both the **functional correctness** and **timing behavior** of the circuit.

PREPARATION FOR LAB 2

- Understand the concept and flow of the synthesis process in digital circuit design.
- Read the server access guide and the user guides for the Xcelium, SimVision and Genus tools.
- Understand how to write and apply TCL (Tool Command Language) scripts to automate the synthesis process in Genus.

REFERENCE

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EXPERIMENT 1

Objective: Investigate the characteristics of standard cells under specific process corners in 45nm technology.

Description: Each standard cell library is characterized at a specific corner, which represents a set of environmental conditions - Process, Voltage, and Temperature (PVT) - under which the cells are simulated and measured. Based on the file name, we can infer the corresponding corner conditions.

For example: the corner of the library *fast_vdd1v0_basicCells_hvt.lib* is:

Process = Fast, Voltage = 1.0 V, Temperature ≈ 25 °C, with High-Vt (HVT) transistor type.

In this experiment, we will investigate the standard cells under eight different process corners in 45nm technology. The conditions for these eight corners are provided in Table 1.

Library Name	Process Corner	VDD (V)	Temperature (°C)	Threshold Type
fast_vdd1v0_basicCells_hvt.lib	Fast (FF)	1.0 V	25 °C (typ.)	HVT (High-Vt)
fast_vdd1v2_basicCells_hvt.lib	Fast (FF)	1.2 V	25 °C (typ.)	HVT (High-Vt)
slow_vdd1v0_basicCells_hvt.lib	Slow (SS)	1.0 V	125 °C (worst case)	HVT (High-Vt)
slow_vdd1v2_basicCells_hvt.lib	Slow (SS)	1.2 V	125 °C (worst case)	HVT (High-Vt)
fast_vdd1v0_basicCells_lvt.lib	Fast (FF)	1.0 V	25 °C (typ.)	LVT (Low-Vt)
fast_vdd1v2_basicCells_lvt.lib	Fast (FF)	1.2 V	25 °C (typ.)	LVT (Low-Vt)
slow_vdd1v0_basicCells_lvt.lib	Slow (SS)	1.0 V	125 °C (worst case)	LVT (Low-Vt)
slow_vdd1v2_basicCells_lvt.lib	Slow (SS)	1.2 V	125 °C (worst case)	LVT (Low-Vt)

Table 1: Operating conditions of the eight analyzed corners.

Instruction:

Students perform the following steps to analyze a single cell. In the following example, we perform the analysis on a AND2 cell.

1. Write a HDL program for a 2-input AND gate. If needed, we can also write the testbench as well. For example:

module nand2 (
  input logic A,
  input logic B,
  output logic Y
);
  assign Y = A & B;
endmodule
	`timescale 1ns/1ps
	module tb_and2;
	  logic A, B;
	  logic Y;
	
	  and2 DUT (.A(A), .B(B), .Y(Y));
	
	  initial begin
	  $dumpfile("and2_wave.vcd");
	  $dumpvars(0, tb_and2);
	
	  A = 0; B = 0; #10;
	  A = 0; B = 1; #10;
	  A = 1; B = 0; #10;
	  A = 1; B = 1; #10;
	  $finish;
	  end
	endmodule

HDL code for cell nand2	**Testbench for cell nand2**

2. Write a TCL script for Cadence Genus to synthesize the and2 cell from the and2.sv file. The script should perform the following steps: read the RTL file (and2.sv), load the standard cell library, perform synthesis, and generate the netlist, reports (power, area), and SDF file.

```
set_db  init_lib_search_path {… }              #Path to library
set_db  init_hdl_search_path {… }              #Path to HDL source code folder
read_libs  slow_vdd1v2_basicCells_lvt.lib   #Read library corner

read_hdl  -sv  and2.sv                         #Read HDL file
elaborate

set_db syn_generic_effort medium
```

theo lý thuyết thì elaborate sẽ đưa về hàm Boolean, nhưng ở đây hàm ý là có nhiều module nối lại với nhau, và kiểm tra có lỗi hay không.

syn_generic mới chuyển đổi thiết kế, ví dụ phép + có thể thành OR hay XOR gì đó.

```
set_db syn_map_effort medium
set_db syn_opt_effort medium

syn_generic
syn_map
#syn_opt                ---> We do not use this command in this experiment

#Generate reports
report_timing  > 02_reports/timing.rpt
report_power   > 02_reports/power.rpt
report_area    > 02_reports/area.rpt
report_qor     > 02_reports/qor.rpt

#Generate netlist and sdf file
write_hdl      > 03_outputs/and2_netlist.sv
write_sdf -timescale ns -nonegchecks -recrem split -edges check_edge -setuphold split > 03_outputs/delay.sdf
```

syn_map=>technology_mapping=>thực hiện gắn với Standard Cells

syn_opt => Thực hiện optimize => Thực hiện ràng buộc thời gian

ghi những report về power, area, delay

3. Students can read the report files (files with the .rpt extension) to extract information about power, cell count, and area utilization.
4. We focus on two files: the netlist file and the SDF file.

The netlist file describes the hardware structure of the circuit after it has been synthesized from the RTL code. It contains a list of logic cells and the connections between them.

```
module and2 (a, b, y);
  input a, b;
  output y;
  wire a, b, y;
  AND2XLLVT g11__2398 (.A(a), .B(b), .Y(y));
endmodule
```

chạy lệnh: genus -f synth.tcl

mở netlist: gui_show

The SDF file (Standard Delay Format) contains information about delays and timing checks of the design after synthesis or layout.

```
(DELAYFILE
  (SDFVERSION "OVI 3.0")
  (DESIGN "and2")
  (DATE "Tue Nov 04 02:10:01 +07 2025")
  (VENDOR "Cadence, Inc.")
  (PROGRAM "Genus(TM) Synthesis Solution")
  (TIMESCALE 1ns)
  (CELL
    (CELLTYPE "AND2XLLVT")
    (INSTANCE g11__2398)
    (DELAY
      (ABSOLUTE
        (PORT A (::0.000))
        (PORT B (::0.000))
        (IOPATH A Y (::0.036) (::0.022))
        (IOPATH B Y (::0.032) (::0.019))
      )
    )
  )
)
```

(TIMESCALE 1ns): The time unit used in the SDF file is 1 nanosecond.

(CELLTYPE "AND2XLLVT"): The name of the cell in the standard cell library (a 2-input AND gate, Low-Vt type).

(INSTANCE g11__2398): The instance name of this cell in the circuit — assigned by the synthesis tool.

ABSOLUTE: Defines absolute delay values, which are the measured delays of the cell under characterized conditions.

(IOPATH A Y (::0.036) (::0.022)):

IOPATH A Y: Represents the signal path from input **A** to output **Y**.

(::0.036) (::0.022):

(min:typ:max)

- The first value (**0.036 ns**) is the delay for a **rising transition** (0 → 1).
- The second value (**0.022 ns**) is the delay for a **falling transition** (1 → 0).

Similarly, **IOPATH B Y** represents the delay from input **B** to output **Y**.

5. After the synthesis process, we perform gate-level simulation to measure the propagation delay. In this step, we must perform the test on the netlist and include the HDL library of the standard cells corresponding to the specific corner (the .v file). The SDF file is included in the testbench to load the delay information into the simulation model.
   Use the command in the testbench to back-annotate delay data:
   $sdf_annotate("../03_outputs/delay.sdf",DUT, , "annotate.log","MAXIMUM");
   A reference testbench is provided in the example below. Note that you should modify the file paths to match the directory structure on your server node.

Định nghĩa GATE_LEVEL để phân biệt file netlist với RTL

dùng thư viện .v để simulate gate_level

```
`timescale 1ns/1ps
`ifdef GATE_LEVEL
        `include ".. /03_outputs/and2_netlist.sv"
        `include ".. /slow_vdd1v2_basicCells_lvt.v"
`else
        `include "../ 00_src/and2.sv"
`endif

module testbench ;
logic A,B;
logic Y;

and2  DUT (.x(A),.y(B),.z(Y));

`ifdef     ANNOTATION
initial begin
        $sdf_annotate("../03_outputs/delay.sdf",DUT, , "annotate.log","MAXIMUM");
```

```
end
`endif

initial begin
        $dumpfile("and2_wave.vcd");
        $dumpvars(0,testbench.DUT);
end
```

viết tb để verify netlist và sdf

```
initial begin
        #0; A=0;B=0;
        #0.1; A=1;B=0;
        #0.1; A=1;B=0;
        #0.1; A=1;B=1;
        #0.1
        $finish;
end

endmodule
```

Requirements:

- Measure the parameters: area, power, and delay for the cells NOT, OR2, NAND2, NOR2, MUX, and FULL ADDER. Each cell must be measured across all 8 corners.

Your report has to show two results:

- For the power and area parameters, it is sufficient to present the results in a tabular format.
- For the delay parameter, in addition to the data table, students are required to present the waveform as proof.

srun -define GATE_LEVEL -define ANNOTATION +access+rw 01_tb/testbench.sv -messages

=> simvision <wave>.vcd &

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EXPERIMENT 2

Objective: Perform gate-level simulation on the netlist after synthesis.

Description: In Experiment 1, we analyzed combinational logic cells. In this experiment, we will examine a more complex combinational circuit.

Given the following code:

```
module alu (input logic [1:0] A,
            input logic [1:0] B,
            input logic [1:0] S,
            output logic [1:0] Y);

always_comb begin
case (S)
2'b00: Y=A+B;
2'b01: Y=A-B;
2'b10: Y=A&B;
2'b11: Y=A|B;
end

endmodule
```

Requirements:

Use the library *slow_vdd1v2_basicCells_hvt.lib* to do the following requirements:

- Synthesize the code in the description with no constraints (do not use file sdc).
- Write a testbench program that applies the input signals of the ALU according to the table below, with each input set applied after a time interval T.

S	A	B
2'b00	2'b01	2'b00
2'b00	2'b11	2'b01
2'b01	2'b11	2'b00
2'b01	2'b11	2'b11

2'b10	2'b11	2'b01
2'b10	2'b01	2'b10
2'b11	2'b11	2'b01
2'b11	2'b01	2'b10

Do this requirement in three cases: T=20, T=100 and T=200

Your report has to show two results:

- The schematic and netlist file after synthesis.
- The waveform result when applying the input signals to the ALU according to the table in all three cases of T. Specify the correct result when no delay is present. For each case, explain why the obtained result appears as it does.

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EXPERIMENT 3

Objective: Investigate the characteristics of standard cells under specific process corners in 45nm technology.

Description: In Experiment 1, we analyzed combinational logic cells. In this experiment, we will examine a sequential cell - a D-FF.

The timing parameters of a DFF in the SDF file are represented as follows:

```
(TIMESCALE   1ns)
(CELL
  (CELLTYPE "DFFHQX1LVT")
  (INSTANCE q_reg)
  (DELAY
    (ABSOLUTE
      (PORT CK (::0.000))
      (PORT D (::0.000))
      (IOPATH CK Q (::0.068) (::0.078))
    )
  )
  (TIMINGCHECK
    (HOLD (negedge D) (posedge CK) (::0.005))
    (HOLD (posedge D) (posedge CK) (::0.000))
    (SETUP (negedge D) (posedge CK) (::0.016))
    (SETUP (posedge D) (posedge CK) (::0.036))
  )
)
```

(IOPATH CK Q (::0.068) (::0.078))

- When a clock edge (CK) occurs, the output Q changes after a certain delay.
- 0.068 ns is the delay from the rising edge of CK to the rising transition of Q.
- 0.078 ns is the delay from the rising edge of CK to the falling transition of Q.

(HOLD (negedge D) (posedge CK) (::0.005)): After the rising edge of **CK**, the input **D** must remain stable for at least **0.005 ns** if **D** has just transitioned low.

(HOLD (posedge D) (posedge CK) (::0.000)): If D has just transitioned high, it must remain stable for 0 ns after the rising edge of CK.

(SETUP (negedge D) (posedge CK) (::0.016)): The input D must be stable for 0.016 ns before the rising edge of CK if D transitions low.

(SETUP (posedge D) (posedge CK) (::0.036)): The input D must be stable for 0.036 ns before the rising edge of CK if D transitions high.

Requirements:

- Measure the parameters: area, power, and delay for the D-FF cell. It must be measured across all 8 corners.
- Verify the setup and hold times of the DFF.

Your report has to show two results:

- For the power and area parameters, it is sufficient to present the results in a tabular format.
- For the delay parameter, in addition to the data table, students are required to present the waveform as proof.

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EXPERIMENT 4

Objective: Perform gate-level simulation on the netlist after synthesis.

Description: In Experiment 1 and 3, we analyzed combinational logic cells and a DFF cell. In this experiment, we will examine a simple squential circuit.

Given the following code:

```
module counter (input logic clk,
                input logic rst,
                output logic [1:0] Q);

always_ff @ (posedge clk) begin
if (rst)
    Q  <=  2'b00;
else
    Q <= Q + 2'b01;
end

endmodule
```

Requirements:

Use the library *slow_vdd1v2_basicCells_hvt.lib* to do the following requirements:

- Synthesize the code in the description with no constraints (do not use file sdc).
- Write a testbench program for an up-counter, where the clock period is T. Do this requirement in three cases: T=50, T=100 and T=200.

Your report has to show two results:

- The schematic and netlist file after synthesis.
- Present the waveform results for all three cases of the clock period T. Identify the correct result when there is no delay, and for each case, explain why the output behaves that way.